MILITARY SPECIFICATION

AMPLIFIERS, SERVOMOTOR GENERAL SPECIFICATION

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the general requirements for transistorized, power amplifiers for driving servomotor control windings either directly or through matching transformers.

1.2 Classification -

1.2.1 Type -

Type 1 - Repairable

Type 2 - Non-Repairable

1.2.2 Class - Amplifiers covered by this specification shall be classed in accordance with MIL-E-5400.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-3965	Capacitors, Fixed, Electrolytic, Tantalum

FSC 5990

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SPECIFICATIONS

Military (Continued)

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-P-7936	Parts and Equipment Procedures for Packaging of
MIL-I-16910	Interference Measurement, Radio, Methods and Limits, 14 Kilocycles to 1000 Megacycles
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-B-22205	Bags, Transparent, Flexible, Heat Sealable for Packaging Applications
MIL-R-22973	Reliability Index Determination for Avionic Equipment Models, General Specification for
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment General Specification for

STANDARDS

Military

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-756	Reliability of Weapons Systems, Procedures for Prediction and Reporting Prediction of

(When requesting any of the applicable documents, refer to both title and number. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer,Naval Supply Depot (Code 1051), 5801 Tabor Avenue,Philadelphia,Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Preproduction - The amplifiers furnished under this specification shall be a product, samples of which have been tested and have passed the preproduction tests specified herein.

3.2 Quality conformance - Each amplifier furnished under this specification shall have demonstrated that it fulfills all of the requirements of this specification by satisfactorily performing all of the Individual tests specified herein, and being one of a lot from which representative units have passed the specified Sampling tests.

3.3 Specification sheets - The individual part requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of conflict between this specification and the specification sheet, the latter shall govern.

3.4 Design and construction -

3.4.1 Compliance with Specification MIL-E-5400 - Amplifiers furnished under this specification shall comply with all requirements of MIL-E-5400 with the following exceptions:

(a) Requirements in conflict with the applicable specification sheets.

(b) Requirements in conflict with this specification.

(c) Requirements which are not applicable, such as those pertaining to materials, parts, or processes not used in the construction of the amplifier.

3.4.2 Adjustments - Unless otherwise specified in the applicable specification sheets, no adjustment for gain or phase shift shall be provided.

3.4.3 Capacitors - Capacitors used throughout the amplifier shall conform to MIL-C-3965.

3.4.4 Semiconductors - Semiconductors used throughout the amplifier shall be silicon, conforming to MIL-S-19500.

3.4.5 Non-magnetic materials - Non-magnetic materials shall be used for all parts of the amplifier except where magnetic properties are essential.

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3.4.6 Magnesium - Magnesium or magnesium alloys in any form shall not be used.

3.4.7 Physical characteristics - The physical characteristics of the amplifier shall be in accordance with the applicable specification sheets.

3.4.8 Electrical characteristics -

3.4.8.1 Quiescent current - The input current drawn by the amplifier when measured in accordance with 4.3.4 shall not exceed the value specified in the applicable specification sheet.

3.4.8.2 Output noise - The output noise level of the amplifier shall be no greater than the maximum specified in the applicable specification sheet when the amplifier is tested in accordance with 4.3.5.

3.4.8.3 Stability - The amplifier shall operate with no extraneous signal in the output when tested in accordance with 4.3.6.

3.4.8.4 Power supply impedance effect - The amplifier shall operate with no extraneous signal in the output when tested in accordance with 4.3.7.

3.4.8.5 High signal voltage input - The dc current drawn by the amplifier and the output voltage shall be as specified in the applicable specification sheet when the amplifier is tested in accordance with 4.3.8.

3.4.8.6 Saturation voltage - The output voltage and dc supply current shall be as specified in the applicable specification sheet when the amplifier is tested in accordance with 4.3.9.

3.4.8.7 Short circuit current - The dc current drawn by the amplifier shall not exceed the value specified in the applicable specification sheet when the amplifier is tested in accordance with 4.3.10.

3.4.8.8 Power supply overvoltage - The signal voltage gain shall remain within the limits specified in the applicable specification sheet when the amplifier is tested in accordance with 4.3.11.

3.4.8.9 Signal input voltage gain - The voltage gain of the amplifier shall be as specified in the applicable specification sheet when the amplifier is tested in accordance with 4.3.12.

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3.4.8.10 Tachometer input voltage gain - Amplifiers having tachometer input terminals shall be tested in accordance with 4.3.13. The voltage gain shall be as specified in the applicable specification sheet.

3.4.8.11 Phase shift - The phase shift through the amplifier shall be within the limits specified in the applicable specification sheet when measured according to 4.3.14.

3.4.8.12 Signal input impedance - The amplifier input impedance at the signal input terminals shall be as specified in the applicable specification sheet when tested in accordance with 4.3.15.

3.4.8.13 Tachometer input impedance - When tachometer input terminals are provided, the amplifier input impedance at the tachometer input terminals shall be as specified in the applicable specification sheet when measured in accordance with 4.3.16.

3.4.8.14 Output impedance - The output impedance of the amplifier when measured in accordance with 4.3.17 shall be as specified in the applicable specification sheets.

3.4.8.15 Harmonic rejection - The amplifier output voltage shall not exceed the voltage specified for each harmonic frequency specified in the applicable specification sheet when measured in accordance with 4.3.18.

3.4.8.16 Amplifier bandwidth - The amplifier bandwidth shall be not less than that specified in the applicable specification sheet when the amplifier is tested in accordance with 4.3.19.

3.4.8.17 Warmup time - When measured in accordance with 4.3.20, the time required for the output voltage to stabilize shall be no greater than the warmup time specified in the applicable specification sheet.

3.4.9 Performance -

3.4.9.1 Temperature and Altitude - The amplifier shall perform as specified in the applicable specification sheet while undergoing the temperature and altitude test of 4.3.21.

3.4.9.2 Vibration - The amplifier shall perform as specified in the applicable specification sheet while undergoing the vibration test of 4.3.22.

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3. 4. 9. 3 Shock - The amplifier shall perform as specified in the applicable specification sheet while undergoing the shock test of 4. 3. 23.

3. 4. 9. 4 Moisture resistance - The amplifier shall perform as specified in the applicable specification sheet upon completion of the moisture resistance test of 4. 3. 24.

3. 4. 9. 5 Salt Spray - The amplifier shall perform as specified in the applicable specification sheet upon completion of the salt spray test of 4. 3. 25. Type I amplifiers shall be disassembled and visually inspected; Type 2 amplifiers shall be inspected without disassembly. There shall be no evidence of corrosion which could in any manner prevent the amplifier from meeting the requirements of this specification.

3. 4. 9. 6 Acceleration - The amplifier shall provide the output specified in the applicable specification sheet when tested in accordance with 4. 3. 28.

3. 4. 9. 7 Radio interference - The amplifier shall be so constructed that radiated and conducted radio noise will not interfere with the effective use of communication or navigation equipment when tested in accordance with 4. 3. 29.

3. 4. 9. 8 Reliability - The amplifier reliability shall be planned, designed, and reported in accordance with MIL-STD-756. The reliability goal, expressed in mean-time-between-failure (MTBF), shall be as specified in the applicable specification sheets. The confidence level shall be 90 percent. To determine the MTBF actually attained, the amplifier shall be tested in accordance with 4. 4. 30. The quantity of amplifiers to be tested shall be as specified in the contract. For preproduction testing the test duration and the maximum number of failures shall be as specified in the contract (see 6. 2). Quality Conformance Inspection for production lots shall be according to MIL-R-23094. The test level and accept-reject criteria shall be as specified in the applicable specification sheet.

3. 4. 10 Identification - The following information shall be legibly and permanently marked on the amplifier case or on a nameplate securely attached thereto:

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U. S. Navy - BuWeps
Amplifier - Servomotor
(Type Designation)
(Specification Sheet No.)
Part No. (Gov't)
Manufacturer's Part No.
Contract or Order No.
Manufacturer's Name or Trademark

3.4.11 Workmanship - Workmanship shall be of the level associated with high grade avionics equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests - The inspection and testing of amplifiers shall be classified as follows:

(a) Preproduction tests.

(b) Quality conformance inspection.

4.2.1 Preproduction tests - The preproduction tests of the amplifier shall consist of the tests listed in Table I.

4.2.1.1 Preproduction test sample - The number of units in the preproduction test sample shall be as specified in the contract or purchase order. The sample shall consist of amplifiers manufactured and packaged in accordance with this specification. The units submitted for preproduction testing shall not have been previously tested except for the Individual tests. The sample shall be forwarded to the testing laboratory designated by the procuring activity (see 6.2).

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TABLE I

Preproduction Tests
(Perform tests in the order listed)

Rough handling	(4.3.1)
Visual inspection	(4.3.2)
Weight	(4.3.3)
Quiescent current	(4.3.4)
Output noise	(4.3.5)
Stability	(4.3.6)
Power supply impedance effects	(4.3.7)
High signal voltage input	(4.3.8)
Saturation voltage	(4.3.9)
Short circuit current	(4.3.10)
Power supply overvoltage	(4.3.11)
Signal input voltage gain	(4.3.12)
Tachometer input voltage gain	(4.3.13)
Phase shift	(4.3.14)
Signal input impedance	(4.3.15)
Tachometer input impedance	(4.3.16)
Output impedance	(4.3.17)
Harmonic rejection	(4.3.18)
Amplifier bandwidth	(4.3.19)
Warmup time	(4.3.20)
Temperature and altitude	(4.3.21)
Vibration	(4.3.22)
Shock	(4.3.23)
Moisture resistance	(4.3.24)
Salt spray	(4.3.25)
Acceleration	(4.3.26)
Radio interference	(4.3.27)
Reliability	(4.3.28)

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4.2.1.2 Preproduction test sample identification - The test sample units shall be plainly identified by durable tags, securely attached, and marked with the information listed below:

Sample for Preproduction Testing

Amplifier, Servomotor (Type as designated in specification sheet)

Submitted by (Manufacturer's name, date) for Preproduction Testing in accordance with MIL-A-81307(WP) under contract or order number (Contract or Order No.)

Manufacturer's Part Number

4.2.2 Quality conformance inspection - The quality conformance inspection of the amplifiers shall consist of Individual tests and Sampling tests.

4.2.2.1 Individual tests - Each amplifier shall be subjected to the tests listed in Table II.

TABLE II

Individual Tests
(Perform tests in the order listed)

Visual inspection	(4.3.2)
Quiescent current	(4.3.4)
Stability	(4.3.6)
High signal voltage input	(4.3.8)
Saturation voltage	(4.3.9)
Signal input voltage gain	(4.3.12)
Tachometer input voltage gain	(4.3.13)
Signal input impedance	(4.3.15)
Output impedance	(4.3.17)

4.2.2.2 Sampling tests - Samples of the product which have passed the Individual tests shall be selected at random in accordance with the following schedule and subjected to the tests listed in Table III.

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Sampling Schedule

Quantity of Amplifiers Offered for Acceptance	Number of Sample Units
First 15	1
Next 25	1
Next 50	1
Next 100	1
Each additional 200 or fraction thereof	1

Failure of any sample shall be cause for the rejection of the quantity offered for acceptance.

TABLE III

Sampling Tests
(Perform tests in the order listed)

	Weight	(4.3.3)
	Short circuit current	(4.3.10)
	Tachometer input voltage gain	(4.3.13)
	Phase shift	(4.3.14)
	Tachometer input impedance	(4.3.16)
	Harmonic rejection	(4.3.18)
	Amplifier bandwidth	(4.3.19)
	Temperature and altitude	(4.3.21)
	Vibration	(4.3.22)
	Shock	(4.3.23)
	Moisture resistance	(4.3.24)
	Salt spray	(4.3.25)
*	Reliability	(4.3.28)

* Destructive test see 6.2

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4.2.3 Test conditions - Unless otherwise specified herein or in the applicable specification sheet, all tests required shall be performed under the following conditions:

Temperature - Room ambient

Atmospheric Pressure - Room ambient

Humidity - Room ambient

Notes:

1. Whenever the amplifier load is used as a component of the amplifier circuit, the effect of the load on the measured parameters shall be taken into consideration. For example: In some amplifiers, the output transistors are supplied with dc through the load. In this case the "output impedance test" would measure the amplifier output impedance in parallel with the load impedance, and the amplifier output impedance would be calculated from the combination.

4.3 Test methods -

4.3.1 Rough handling - Amplifiers submitted for sampling tests, shall be packaged and packed in a manner typical of the amplifiers to be delivered under contract or order. Rough handling tests of packaging and packing shall be performed on the unopened packages in accordance with MIL-P-7936.

4.3.2 Visual inspection - The amplifier shall be visually inspected to determine conformance to this specification and the applicable detail specification. Particular attention shall be given to the requirements concerning dimensions affecting interchangeability, materials, marking, finish, and workmanship.

4.3.3 Weight - The weight shall be determined within ± one percent.

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4.3.4 Quiescent dc current - Connect the amplifier as shown in Figure 1. Adjust the dc power supply to provide the nominal amplifier supply voltage. Measure the current drawn by the amplifier.

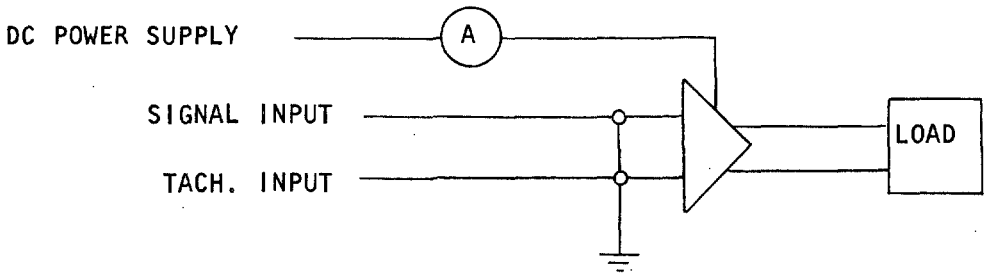


Figure 1. Circuit for quiescent current measurement.

4.3.5 Output noise - Connect the amplifier and the load specified in the applicable specification sheet as shown in Figure 2. The voltage read on the voltmeter is the output noise level.

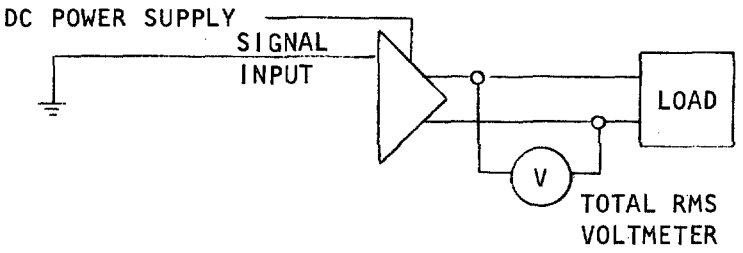


Figure 2. Circuit for output noise measurement.

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4.3.6 Stability - Connect the amplifier as shown in Figure 3. Sweep the signal source throughout the voltage range specified in the applicable specification sheet for at least five sweeps. Inspect the amplifier output waveform for extraneous signal.

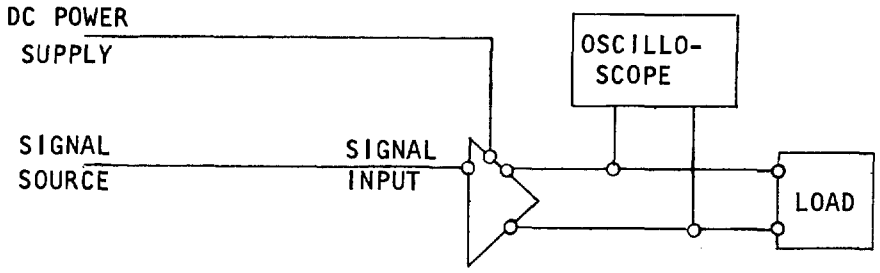


Figure 3. Circuit for observing amplifier stability

4.3.7 Power supply impedance effect - Connect the amplifier as shown in Figure 4, using the values for R1 & R2 specified in the applicable specification sheet. Inspect the amplifier output waveform for extraneous signal.

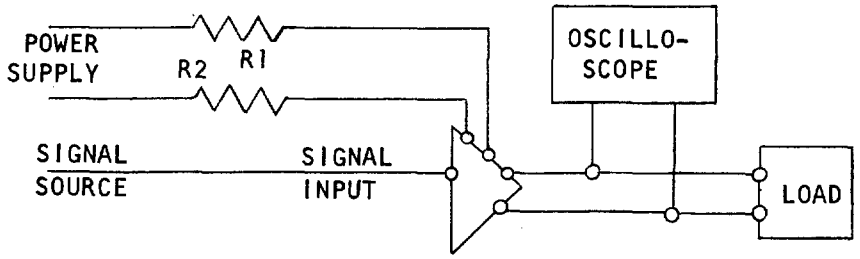


Figure 4. Circuit for observing power supply impedance effect.

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4.3.8 High signal voltage input - Connect the amplifier as shown in Figure 5. Apply the signal voltage specified in the applicable specification sheet. The voltmeter will read the output voltage.

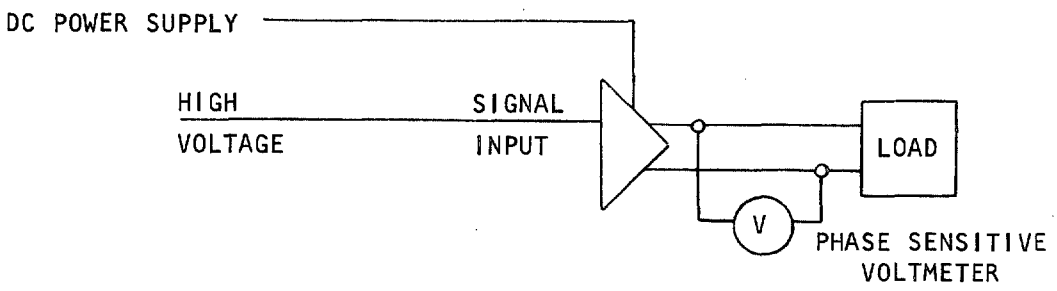


Figure 5. Circuit for high signal voltage input measurement.

4.3.9 Saturation voltage - Connect the amplifier as shown in Figure 6. Adjust the signal source to the voltage specified in the applicable specification sheet. Measure the dc supply current and the amplifier output voltage.

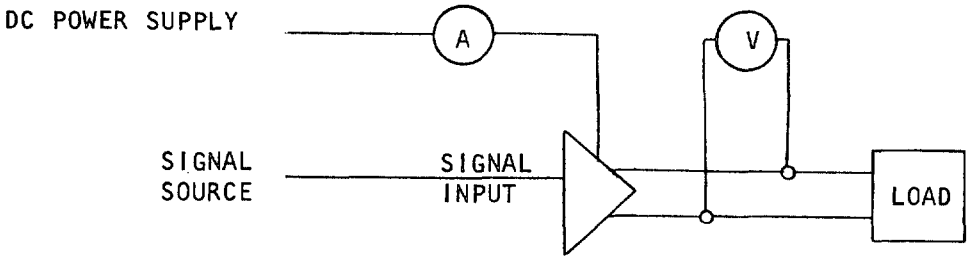


Figure 6. Circuit for amplifier measurements at saturation.

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4.3.10 Short circuit current - Connect the amplifier as shown in Figure 7. Adjust the signal source to the voltage specified in the applicable specification sheet. Measure the dc supply current drawn by the amplifier.

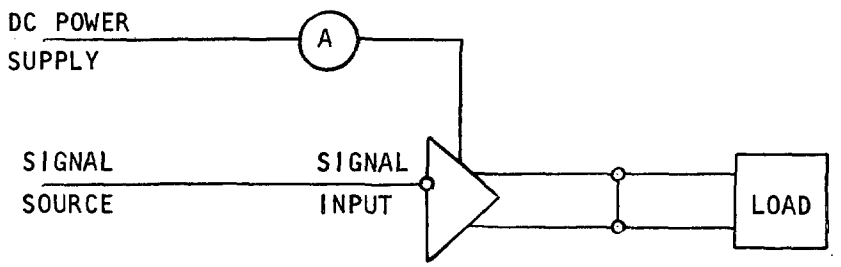


Figure 7. Circuit for short circuit current measurement.

4.3.11 Power supply overvoltage - Connect the amplifier as shown in Figure 8. Adjust the signal source to the voltage specified in the applicable specification sheet. Apply the power supply voltage specified for the length of time specified in the applicable specification sheet. Upon completion of the overvoltage condition, measure the signal input voltage gain as specified in 4.3.12.

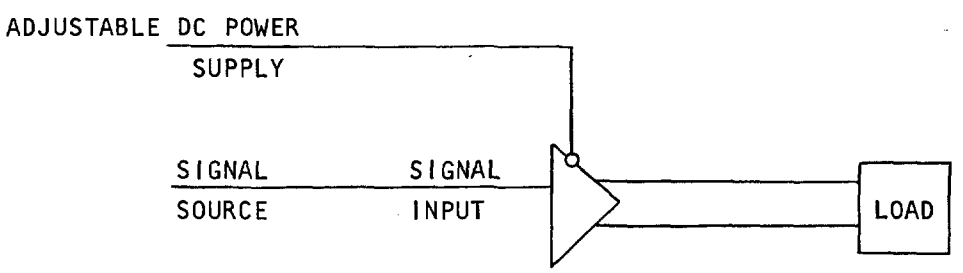


Figure 8. Circuit for power supply overvoltage.

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4.3.12 Signal input voltage gain - Connect the amplifier as shown in Figure 9. Set the signal source to obtain the amplifier output voltage specified in the applicable specification sheet. Measure the input voltage and calculate the voltage gain.

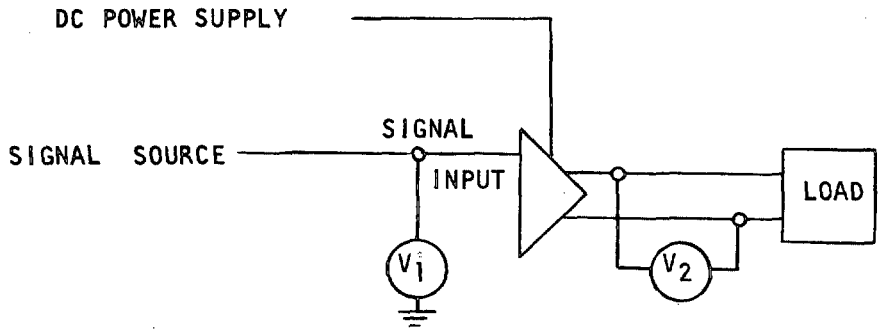


Figure 9. Circuit for signal input voltage gain measurement.

4.3.13 Tachometer input voltage gain - Connect the amplifier as shown in Figure 10. Adjust the signal source to provide the output voltage specified in the applicable specification sheet. Measure the input voltage and calculate the voltage gain.

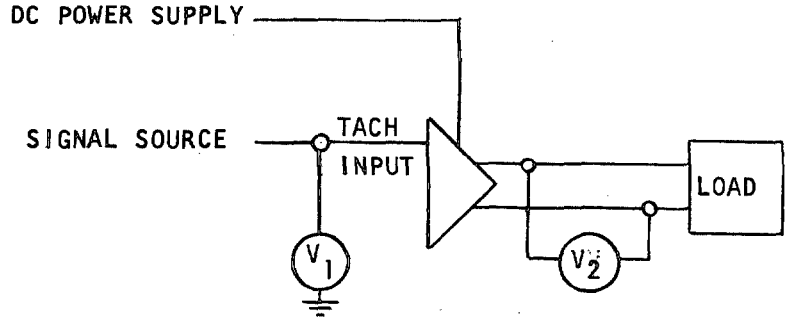


Figure 10. Circuit for tachometer input voltage gain measurements.

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4.3.14 Phase shift - Connect the amplifier as shown in Figure 11. Adjust the signal source to the voltage value specified in the applicable specification sheet. Measure the fundamental and quadrature voltages on the phase sensitive voltmeters. From these values calculate the internal phase shift.

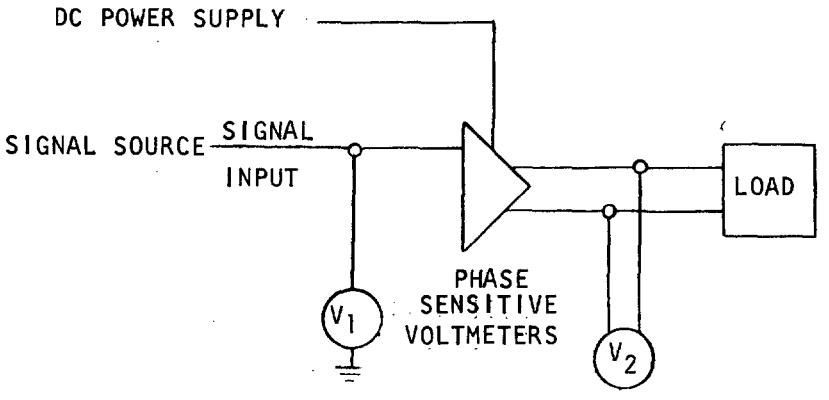


Figure 11. Circuit for phase shift measurement.

4.3.15 Signal input impedance - Connect the amplifier as shown in Figure 12. Adjust the signal source to obtain the output voltage specified in the applicable specification sheet. Adjust R_V until actuating the pushbutton switch produces no change in the amplifier output voltage. The value of R_V is equal to the input impedance.

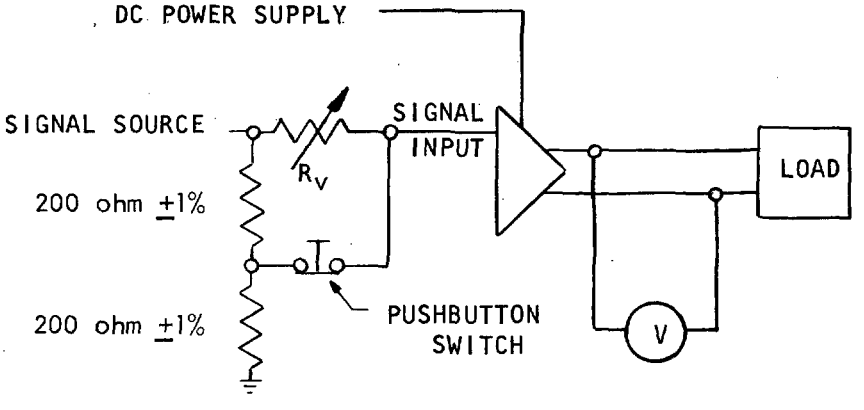


Figure 12. Circuit for signal input impedance measurement.

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4.3.16 Tachometer input impedance - Connect the amplifier and the load specified in the applicable specification sheet as shown in Figure 13. Adjust the signal source to obtain the voltage and frequency specified under this paragraph number in Table I of the applicable specification sheet at the amplifier output. Adjust R_V until actuating the pushbutton produces no change in the amplifier output voltage. The value of R_V is equal to the input impedance.

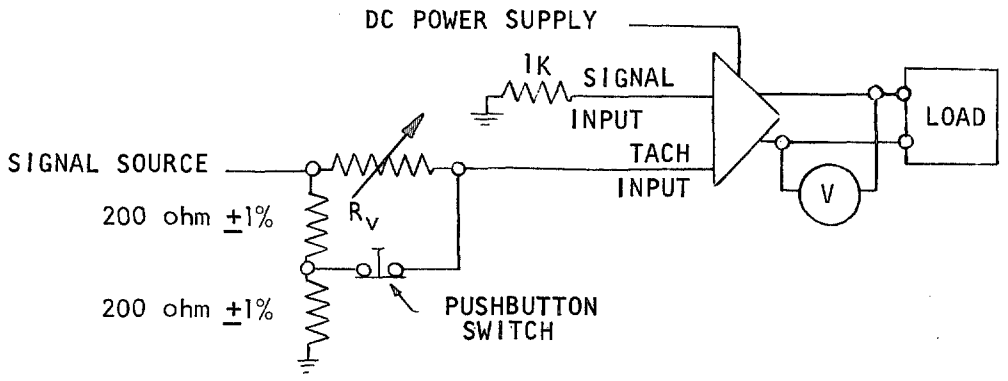


Figure 13. Circuit for tachometer input impedance measurement.

4.3.17 Output impedance - Connect the amplifier and the load specified in the applicable specification sheet as shown in Figure 14. If a center-tapped load is specified, the center-tap shall be connected as specified in the applicable specification sheet. Set resistor R_1 to 10,000 ohms before applying power to the test circuit. Adjust the variable autotransformer T_1 until the a.c. voltmeter, V_1, reads 100 volts. Readjust R_1 until the a.c. voltmeter, V_2, reads 2.0 volts. Calculate R_p, the impedance of the parallel combination of load impedance and the amplifier output impedance, using the equation: $R_p = .02\ R_1$. Calculate R_o, the amplifier output impedance, using the equation:

$$R_o = \frac{R_P\ R_L}{R_L - R_P}$$ where:

R_o = Amplifier output impedance
R_P = Impedance of amplifier output and load in parallel
R_L = Load impedance

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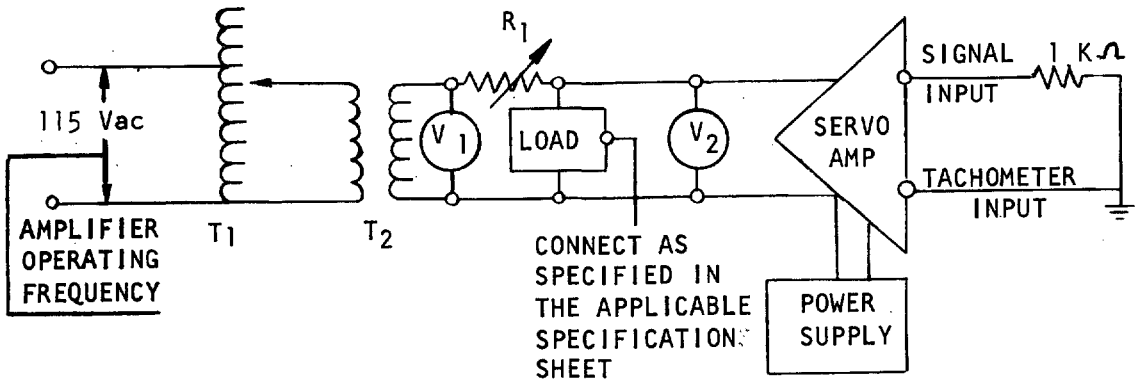


Figure 14. Circuit for amplifier output impedance measurement.

4.3.18 Harmonic rejection - Connect the amplifier as shown in Figure 15. Adjust the signal source to the input voltage and frequency specified in the applicable specification sheet. Measure the output voltage. Readjust the signal source to the same input voltage amplitude, but at the amplifier operating frequency. Measure the output voltage. Calculate the harmonic voltage rejection ratio in db.

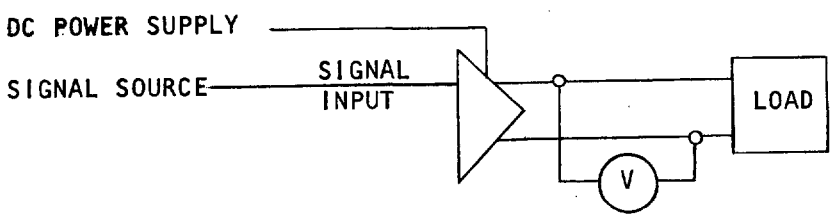


Figure 15. Circuit for harmonic rejection and amplifier bandwidth measurements.

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4.3.19 Amplifier bandwidth - Connect the amplifier as shown in Figure 15. Adjust the signal source to the rated frequency of the amplifier and to the signal input voltage specified in Table I of the applicable specification sheet. Measure the amplifier output voltage. Adjust the signal frequency downward until the amplifier output voltage is 70.7% of the original value. This frequency is the lower limit of the bandwidth. Adjust the signal frequency upward, past the rated frequency, until the amplifier voltage is again 70.7% of the original value. This frequency is the upper limit of the bandwidth.

4.3.20 Warmup time - Connect the amplifier and the load specified in the applicable specification sheet as shown in Figure 16. Adjust the signal source to the voltage and frequency specified in the applicable specification sheet. Depress the pushbutton and measure the time required for the output voltage to settle within five percent of its final value.

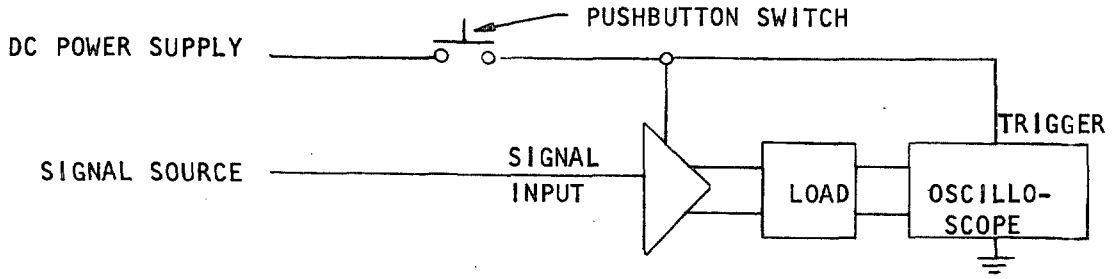


Figure 16. Circuit for warmup time measurement.

4.3.21 Temperature and altitude - The amplifier shall be subjected to the following temperature - altitude tests and in accordance with the applicable specification sheet. Upon completion of the temperature and altitude tests the amplifier shall be removed from the test chamber and allowed to remain in room ambient conditions for at least four hours before undergoing further tests.

4.3.21.1 Low temperature-altitude - The amplifier shall be placed in a test chamber. The temperature of the test chamber shall be reduced to and maintained at the temperature specified in the applicable specification sheet. After the unenergized amplifier has remained at this temperature for at least four hours, the pressure shall be reduced to and maintained at that pressure simulating the altitude specified in the applicable specification sheet. While at this temperature and pressure the

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amplifier shall be subjected to the tests specified in the applicable specification sheet.

4.3.21.2 High temperature-altitude - The temperature shall be increased to and maintained at the temperature specified in the applicable specification sheet. The pressure in the test chamber shall be equalized to room ambient during the temperature change cycle. After the unenergized amplifier has remained at this temperature for at least four hours, the pressure of the chamber shall be reduced to and maintained at the pressure simulating the altitude specified in the applicable specification sheet. While at this temperature and pressure the amplifier shall be subjected to the tests specified in the applicable specification sheet.

4.3.21.3 Low temperature storage - The test chamber shall be reduced to the low-temperature storage temperature specified in the applicable specification sheet. The amplifier shall remain at this temperature for at least 24 hours. The temperature shall then be raised to and maintained at the low operating temperature specified in the applicable specification sheet. After the amplifier has stabilized at this temperature, it shall be subjected to the tests specified in the applicable specification sheet.

4.3.21.4 High temperature storage - The temperature of the test chamber shall be raised to and maintained at the high temperature storage temperature specified in the applicable specification sheet. The amplifier shall remain at this temperature for at least 24 hours. The temperature shall then be set to the high operating temperature specified in the applicable specification sheet. After the amplifier has stabilized at this temperature, it shall be subjected to the tests specified in the applicable specification sheet.

4.3.22 Vibration - The amplifier shall be rigidly mounted on the vibration table or test fixture, using its normal mounting means. While being subjected to the vibration test of MIL-STD-202, Test Method 204, Condition B the amplifier shall perform the tests specified in the applicable specification sheet.

4.3.23 Shock - The amplifier shall be securely fastened to a rigid test fixture, utilizing its normal mounting means, and the test fixture rigidly mounted to the shock test device. The amplifier shall be subjected to 30 impacts of an acceleration of 50 G's, of 11 ± 1 milliseconds time duration. The shock pulse shall be a half sine wave with not more than 10% of the peak pulse distortion at any point. The amplifier shall be subjected to five blows in each direction along the three mutually perpendicular axes (30 blows). While undergoing the shock test, the amplifier shall be subjected to the tests specified in the applicable specification sheet.

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4.3.24 Moisture resistance - The amplifier shall be subjected to the moisture resistance test in accordance with MIL-STD-202, Test Method 106. Tests to be performed in conjunction with the moisture resistance test shall be as specified in the applicable specification sheet.

4.3.25 Salt spray - When required by the applicable specification sheet, the amplifier shall be subjected to the salt spray test in accordance with MIL-STD-202, Test Method 101, Condition B, with a salt-solution concentration of five percent. Tests to be performed in conjunction with the salt spray test shall be as specified in the applicable specification sheet.

4.3.26 Acceleration test - The acceleration test shall be performed as follows:

(a) Connect the amplifier as shown in Figure 9.
(b) Adjust the signal generator to produce the amplifier output voltage, specified in the applicable specification sheet.
(c) Perform the test in accordance with MIL-STD-202, Method 212, Condition A at the 100 G level.
(d) Read the input voltage, and calculate the voltage gain.

4.3.27 Radio interference test - Radiated and conducted radio frequency noise shall be within the limits specified in MIL-I-16910 when tested according to the applicable procedure specified herein.

4.3.28 Reliability test - The reliability test shall be performed as follows:

(a) Connect the amplifier as shown in Figure 9.
(b) The amplifier shall be tested in accordance with MIL-R-22973. The test cycle shall be in accordance with the applicable specification sheets. This cycle shall be repeated until the completion of at least the number of hours specified in the specification sheets. Time counted toward the reliability test shall be the sum of only the power "ON" periods.
(c) At the end of each four cycles, with the power "ON", the voltage gain at the output voltage specified in the applicable specification sheet shall be measured in accordance with 4.3.12.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5.1.1 Cleaning, drying and preservation - Not applicable.

5.1.2 Unit packaging - Unless otherwise specified by contract or order, each amplifier shall be individually packaged in a hermetically sealed container in accordance with MIL-P-7936 with the cushioning necessary to provide a tight package. The cushioning material shall have a pH value between 6.0 and 8.0, and shall not produce corrosive or deleterious fumes or vapors which could act on or attack the amplifier.

5.1.3 Hardware - All hardware required by the specification sheet; such as, terminal lugs, terminal screws, etc. shall be packaged in a transparent bag conforming to MIL-B-22205, Type III. The hardware shall be included in the container described in 5.1.2.

5.2 Packing - Unless otherwise specified by contract or order, amplifiers shall be packed in accordance with MIL-P-7936, Level A, B, or C as specified by the procuring activity. (See 6.2)

5.3 Marking - The nomenclature for shipment and storage shall be as specified in the applicable specification sheets.

6. NOTES

6.1 Intended use - The amplifiers covered by this specification are used to supply the power necessary to drive the servomotors found in integrated instrumentation systems, navigational equipments, fire control systems, and control surface systems in shipboard, airborne, and missile applications.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, date, and number of this specification.

(b) Slant number of applicable specification sheets.

(c) Nomenclature of item.

(d) Number of units in preproduction test sample.

(e) Preproduction test laboratory.

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(f) Number of units to be used in the reliability test.

(g) Number of testing hours for the preproduction reliability test.

(h) Number of failures allowed in the preproduction reliability test.

(j) Level of packing.

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractors Data Requirement List) which shall be attached to and made a part of the contract or order. NavWeps Form 4200/15 (Drawings Lists and Specification Requirements) shall be attached where applicable.

6.3.1 Reliability - Reliability reports in accordance with MIL-STD-756.

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